

12/03/26

ASSIGNMENT-01

EEF207

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Question 01:

Given $R_S = 100\text{k}\Omega$

$A_1 = 10 \left(\frac{V}{V}\right)$

$A_2 = 100 \left(\frac{V}{V}\right)$

$A_3 = 1 \left(\frac{V}{V}\right)$

$R_{i/p(1)} = 1\text{M}\Omega$

$R_{i/p(2)} = 100\text{k}\Omega$

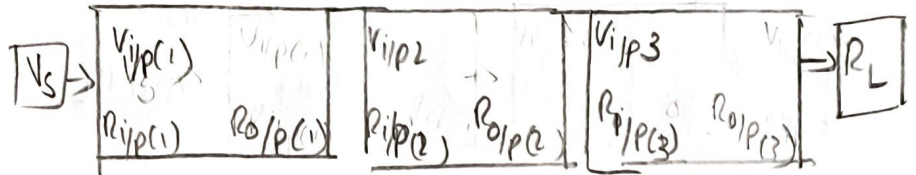
$R_{i/p(3)} = 10\text{k}\Omega$

$R_{o/p(1)} = 1\text{k}\Omega$

$R_{o/p(2)} = 1\text{k}\Omega$

$R_{o/p(3)} = 10\Omega$

$R_L = 100\Omega$



a) overall gain (voltage)

$$A_v = \frac{V_{i/p(1)}}{V_s} \times A_1' \times A_2' \times A_3' = \frac{V_L}{V_s}$$

$$V_{i/p} = V_s \left[\frac{R_{i/p(1)}}{R_{i/p(1)} + R_S} \right]$$

$$\Rightarrow \frac{V_{i/p(1)}}{V_s} = \frac{R_{i/p(1)}}{R_{i/p(1)} + R_S} = \frac{1 \times 10^6}{10^6 + 10^5}$$

$$= \frac{10^6}{10^6 + 0.1 \times 10^6} = \frac{1}{1.1} = 0.909 \left(\frac{V}{V}\right)$$

$$A_v = 0.909 \times 9.909 \times 90.9 \times 0.9090$$

$$A_v = 744.25 \left(\frac{V}{V}\right)$$

$$\text{Current gain: } A_i = \frac{i_L}{i_S} = \frac{\frac{V_{i/p(1)}}{R_{i/p(1)}}}{\frac{V_s}{R_S}} = \frac{0.909 V_s}{\frac{1 \times 10^6}{100 \times 10^3}}$$

$$A_i = 0.0909 \left(\frac{A}{A}\right)$$

$$\text{Power gain: } A_p = A_v A_i = 744.25 \times 0.0909$$

$$= 67.6 \left(\frac{W}{W}\right)$$

$$A_1' = A_1 \times \frac{R_{i/p(2)}}{R_{o/p(1)} + R_{i/p(2)}}$$

$$= \frac{10^3 \times 10^3}{10^3 [1 + 100]} = \frac{10^3}{101} = 9.909 \frac{V}{V}$$

$$A_1' = 9.909 \frac{V}{V}$$

$$A_2' = A_2 \times \frac{R_{i/p(3)}}{R_{o/p(2)} + R_{i/p(3)}}$$

$$= \frac{100 \times 10 \times 10^3}{1 \times 10^3 + 10 \times 10^3} = \frac{10^3}{11}$$

$$A_2' = 90.9 \left(\frac{V}{V}\right)$$

$$A_3' = A_3 \times \frac{R_L}{R_{o/p(3)} + R_L}$$

$$= \frac{1 \times 100}{10 + 100} = 0.909 \frac{V}{V}$$

2) Given

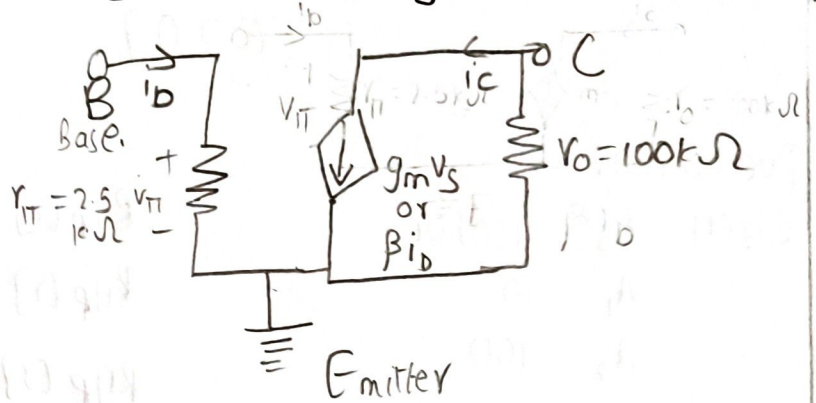
$$R_S = 5k\Omega$$

$$r_{\pi} = 2.5k\Omega$$

$$g_m = 40m\frac{A}{V} = 0.004\left(\frac{A}{V}\right)$$

$$r_o = 100k\Omega$$

$$R_L = 5k\Omega$$

a) small signal hybrid π model

$$b) A_v = \frac{V_o}{V_i}$$

$$V_{\pi} = V_{i/p} \left[\frac{r_{\pi}}{R_S + r_{\pi}} \right]$$

$$\beta = V_{\pi} g_m$$

$$R_{out} = r_o || R_L = \frac{r_o R_L}{r_o + R_L}$$

$$V_o = -\beta R_{out}$$

derivation

$$V_{o/p} = -\beta R_{out}$$

$$= -V_{\pi} g_m R_{out}$$

$$= -V_{i/p} \left[\frac{r_{\pi}}{R_S + r_{\pi}} \right] g_m \left[\frac{r_o R_L}{r_o + R_L} \right]$$

$$A_v = \frac{V_{o/p}}{V_{i/p}} = - \left[\frac{2.5 \times 10^3}{5 \times 10^3 + 2.5 \times 10^3} \right] [0.004] \left[\frac{100 \times 10^3}{100 \times 10^3 + 5 \times 10^3} \right]$$

$$= -[0.333][0.004][0.952]$$

$$A_v = -1.26 \times 10^{-3} \left[\frac{V}{V} \right]$$

$$c) r_o \rightarrow \infty$$

$$r_o || R_L = R_L$$

$$A_v = -g_m R_L \left[\frac{r_{\pi}}{R_S + r_{\pi}} \right]$$

$$= -[0.004][5 \times 10^3] \left[\frac{2.5}{7.5} \right]$$

$$= -[0.004][5 \times 10^3][0.33]$$

$$= 6.66 \times 10^3 \times 10^{-3}$$

$$A_v = 6.6 \left(\frac{V}{V} \right)$$